WORKED SOLUTION

STEP III 1993, Question 13

Mechanics • Variable Mass Chain

STEP III 1993, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/93-s3-q13

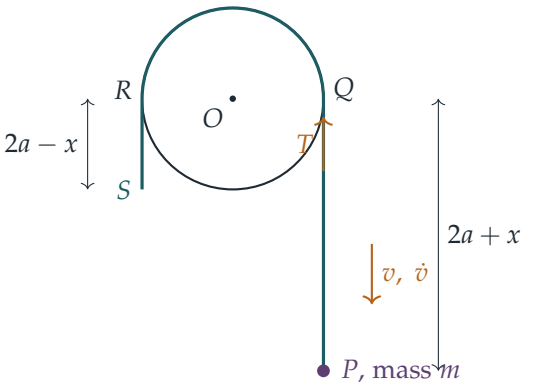
The key idea

Because the chain cannot slip, the disc, the whole chain and the particle all move at the same speed: there is a single unknown, v. The kinetic energy is then $\frac{1}{2}(\text{effective mass})v^2$. In the potential energy only the two hanging pieces change, since the arc over the top just keeps being replaced by identical chain. For the tension, look at the moving piece PQ together with the particle. Then think hard about how fast the chain that joins it at Q is already moving.

Solution

Model

The disc is a uniform lamina turning about a smooth fixed axis through O, with moment of inertia $\frac{1}{2}(4m)a^2 = 2ma^2$. The chain is uniform with mass m/a per unit length, is inextensible, and does not slip on the rim. The particle is a point mass. When P has descended x, the right-hand piece QP has length $2a + x$, the left-hand piece RS has length $2a - x$, and the arc RQ is a semicircle of length πa. Every point of the chain moves at speed v, and since there is no slipping the rim does too, so the disc's angular speed is $\omega = v/a$.



The speed

Kinetic energy. The disc contributes $\frac{1}{2}(2ma^2)(v/a)^2 = mv^2$, the chain $\frac{1}{2}(4+\pi)mv^2$, and the particle $\frac{1}{2}mv^2$. So

$$\text{KE} = \tfrac{1}{2}mv^2(2+4+\pi+1) = \tfrac{1}{2}(\pi+7)mv^2.$$

Potential energy. The arc RQ is always the same semicircle of chain, so its potential energy never changes. Take the level of O as zero. A hanging piece of length ℓ below that level has mass $m\ell/a$ and its centre of mass at depth $\frac{1}{2}\ell$, so its potential energy is $-\dfrac{mg\ell^2}{2a}$. The loss of potential energy of the two hanging pieces is therefore

$$\frac{mg}{2a}\left[(2a+x)^2+(2a-x)^2-(2a)^2-(2a)^2\right] = \frac{mg}{2a}\cdot 2x^2 = \frac{mgx^2}{a}.$$

The particle loses mgx.

Energy. The axis is smooth and the rough contact does no work, because there is no slipping. So the energy is conserved, and starting from rest

$$\tfrac{1}{2}(\pi+7)mv^2 = \frac{mgx^2}{a} + mgx \quad\Longrightarrow\quad (\pi+7)av^2 = 2g(x^2+ax),$$

as required.

The acceleration

Differentiating with respect to time and using $\dot{x} = v$:

$$2(\pi+7)av\dot{v} = 2g(2x+a)v \quad\Longrightarrow\quad \dot{v} = \frac{g(2x+a)}{(\pi+7)a} \qquad (v \neq 0).$$

S reaches R when $x = 2a$, and then

$$\dot{v} = \frac{5g}{\pi+7}, \qquad v^2 = \frac{2g(4a^2+2a^2)}{(\pi+7)a} = \frac{12ga}{\pi+7}.$$

The tension at Q

Let T be the tension at Q. Our system is the particle together with the chain hanging below Q. Its mass is

$$M = \frac{m}{a}(2a + x) + m,$$

which increases at the rate $\dot{M} = \dfrac{m}{a}v$ as chain comes off the disc. The external forces on it are its weight Mg downwards and T upwards. All of it moves downwards at speed v.

Ask yourself How fast is the chain that is joining PQ moving just before it joins?

At Q the rim is vertical, so the chain arriving there is already moving vertically downwards at speed v. It brings its momentum $v\,\delta M$ with it. The variable-mass equation (rate of change of momentum = force + momentum brought in per unit time) is then

$$\frac{\mathrm{d}}{\mathrm{d}t}(Mv) = Mg - T + v\dot{M}, \qquad \text{that is,} \qquad M\dot{v} = Mg - T.$$

When S reaches R, $M = 5m$, so

$$T = 5m\Big(g - \frac{5g}{\pi + 7}\Big) = \frac{5(\pi + 2)}{\pi + 7}\,mg.$$

A cross-check by moments. Take the disc and the arc RQ of chain together. The axle force acts through O, the arc's weight has zero total moment about O (the arc is symmetric about the vertical through O), and the friction between chain and disc is internal. Every particle of the arc moves at speed v at distance a from O. So the angular momentum is $(2ma^2 + \pi ma^2)(v/a)$, and

$$(T - T_R)a = (2 + \pi)ma\,\dot{v}.$$

When S reaches R the left-hand piece has shrunk to nothing, so $T_R = 0$. This gives $T = (2 + \pi)m \cdot \dfrac{5g}{\pi + 7} = \dfrac{5(\pi + 2)}{\pi + 7}mg$ again.

The printed answer. The paper's value $\dfrac{5\pi - 2}{\pi + 7}mg$ is exactly what we get by writing $\dfrac{\mathrm{d}}{\mathrm{d}t}(Mv) = Mg - T$, which treats the chain joining PQ as if it arrived at rest:

$$T = M(g - \dot{v}) - \frac{m}{a}v^2 = \frac{5(\pi + 2)}{\pi + 7}mg - \frac{12}{\pi + 7}mg = \frac{5\pi - 2}{\pi + 7}mg.$$

That is the right equation for a raindrop picking up stationary mist. It is not right here, because the chain coming round the rim is already moving with PQ. Both of our methods agree on $\dfrac{5(\pi + 2)}{\pi + 7}mg$, so we believe the printed answer contains this error. If you need the printed value, use the "joins at rest" equation, but know that it undercounts T by mv^2/a.

Checks

- At $x = 0$ the formula gives $v = 0$ and $\dot{v} = g/(\pi + 7)$. The out-of-balance force is just the particle's weight mg, acting on effective mass $(\pi + 7)m$.
- Stepping the equation of motion $(\pi + 7)m\dot{v} = mg(2x + a)/a$ numerically from rest reproduces $(\pi + 7)av^2 = 2g(x^2 + ax)$ to within 10^{-11}. With $m = a = 1$ and $g = 9.81$, the momentum method and the moments method both give $T/mg \approx 2.5349 = 5(\pi + 2)/(\pi + 7)$. The "joins at rest" version gives $1.3517 = (5\pi - 2)/(\pi + 7)$.
- $T < 5mg$, as it must be, since the hanging mass $5m$ is accelerating downwards.

Common slips

- Forgetting the disc's kinetic energy, or using ma^2 instead of $\frac{1}{2}(4m)a^2$ for its moment of inertia.
- Including the arc of chain in the change of potential energy. Its shape and position never change.
- Using $\frac{\mathrm{d}}{\mathrm{d}t}(Mv) = F$ without asking what velocity the incoming mass already had.

Beyond STEP

Variable-mass problems are really about choosing a system and keeping track of the momentum that crosses its boundary. That is the idea of a control volume, which underlies rocket motion and much of fluid mechanics. The common error here, assuming $F = \frac{\mathrm{d}}{\mathrm{d}t}(mv)$ holds for any changing mass, is a famous trap: in general that equation isn't even independent of the frame of reference. Kleppner and Kolenkow, *An Introduction to Mechanics*, treat systems of variable mass carefully.

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